Munireach Nannory

PhD Candidate | FEA and Machine Learning | LLM Agents for Autonomous Experimentation

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PROFESSIONAL SUMMARY

PhD researcher in FEA and machine learning with ~3 years of experience building predictive nonlinear Abaqus/CAE simulations, simulation-driven ML models, and AI agents for autonomous experimentation. Built an end-to-end Python and MATLAB workflow on a Linux cluster that executed **100,000+ predictive FEA simulations** and saved **~8,000 hours** of human interaction. Developed a **governed LLM multi-agent system** that operates a commercial nanoindenter through its closed vendor software, running a **~3-hour autonomous campaign** after a single human authorization. First-author paper in *J. Mater. Res. Technol.* (2026) and co-author in *Materials & Design* (2026). Clear communicator who turns simulation, ML, and agent results into decision-ready guidance.

AREAS OF EXPERTISE

PROFESSIONAL SKILLS Predictive Nonlinear FEA (contact, large deformation) | Elastic-Plastic Material Modeling | Load Case and BC Definition | Meshing Strategy and Quality | Sensitivity and Parametric Studies | Post-Processing and Interpretation | V&V and Test Correlation | Cross-Functional Collaboration | Technical Documentation

TECHNICAL SKILLS

Machine Learning (Simulation-driven): 1D CNN | Neural Networks | Deep Kernel Learning (DKL) | Hyperparameter Tuning | Uncertainty-Aware Modeling | Attention Models | Symbolic Regression (PySR)

Modeling and Simulation: Abaqus/CAE | Nonlinear FEA | Contact Modeling | Elastic-Plasticity

Automation and Data: Python | MATLAB | Git/GitHub | Linux (Cluster) | Batch Workflows | Standardized Post-Processing | Dataset Generation

AI Agents and Autonomy: LLM Multi-Agent Systems | Vision-Language Models | Bayesian Optimization (GP, EI) | GUI Automation and OCR | Rule-Based Guardrails | Hugging Face

EDUCATION

- **PhD, Civil and Environmental Engineering** | University of Tennessee, Knoxville | May 2024 - Present
  Focus: Finite Element Method, nanoindentation simulation, automation, machine learning, deep learning, LLM agents for autonomous experimentation
  Honors: Student of the Year, University of Tennessee, Knoxville (2025)
  GPA: 3.97
- **MS, Civil and Environmental Engineering** | University of Tennessee, Knoxville | Aug 2023 - May 2025
- **BS, Civil Engineering** | Pañ̃āsāstra University of Cambodia | Nov 2016 - Nov 2022
  Honors: Top ranked scholar (Cohorts 50 and 51)
- **BA, Teaching English as a Foreign Language** | Royal University of Phnom Penh | Feb 2017 - Nov 2022

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (NSF funded), University of Tennessee, Knoxville | Knoxville, TN |Aug 2023 - Present

- Developed predictive nonlinear Abaqus/CAE models (contact, large deformation, elastic-plasticity) for design-relevant load cases with verification-minded modeling practices.
- Automated 100,000+ Abaqus simulations, saving 8,000 hours of human interaction.
- Trained simulation-driven ML models (neural networks, Deep Kernel Learning) to learn input-to-response mappings for rapid screening of risk drivers and design trade-offs.
- Built and deployed a governed LLM multi-agent system that runs closed-loop nanoindentation campaigns on a commercial KLA iMicro, from a plain-English objective to vendor-reported hardness and modulus.
- Performed sensitivity studies and interpolation-based analysis to quantify dominant drivers, and communicated results through technical documentation and decision-ready summaries.
- Contributor to the NSF ATHENA Programmable Cloud Laboratory testbed (Working Group 4: Materials Libraries and Adaptive Testing), focused on agentic nanoindentation and programmatic instrument control.

Selected Projects

Governed Autonomous Nanoindentation with LLM Multi-Agent Control | 2026 - Present

- Architected an orchestrator and six specialist agent roles (image analysis, literature, hypothesis, planning, measurement strategy, measurement critic) on open-weight LLMs (Qwen2.5-72B-Instruct, Qwen2.5-VL-72B) that compile a plain-English objective into weighted property goals, an indent budget, and machine-checkable

acceptance criteria.

- Operated a closed commercial instrument (KLA iMicro/inForce 1000) with no vendor API through a 17-command allow-listed interface built on GUI automation and OCR; a Gaussian-process surrogate with expected improvement selects each next indentation site.
- Designed governance so a deterministic rule engine, not the LLM, holds stop and continue authority, with two-phase human approval, one-time authorization for irreversible indents, exactly-once execution, and provenance logs; the LLM never overrode it in 63 logged decisions.
- Closed the loop in all 10 hardware campaigns (84 indents). In the flagship run, the agents executed rounds 2-10 over ~3 hours with no person choosing a site and relocated ~200 µm to the hardest phase measured (Fe-rich steel: 4.3 GPa hardness, 189 GPa modulus) without access to the EDS map.

FEM + 1D CNN Substrate Correction for Thin-Film Nanoindentation | Nov 2024 - Present

- Generated paired thin-film and bulk curves via FEM sweeps over E, σ_y, and strength/hardening parameters; resampled to a fixed 200-point normalized depth grid with additional input channels.
- Trained a 1D CNN (Huber loss, noise augmentation) to predict bulk response from thin-film indentation curves.
- Achieved <2% validation MAE on held-out simulations; reduced experimental copper thin-film indentation error by ~60–80% versus uncorrected measurements (with uncertainty bands).
- Enabled improved extraction of bulk-consistent properties including E, σ_y, and ultimate strength after correction.

Substrate-Effect Sensitivity and Regime Mapping (FEM Automation + Data-Driven Guidance) | Jul 2024 - Present

- Automated high-throughput simulation campaigns to produce response maps, regime boundaries, and depth-binned error metrics for reliable decision-making.
- Quantified sensitivity trends to reduce trial-and-error and support simulation-driven planning.
- Standardized mesh quality and modeling assumptions across batches and documented recommended meshing practices to balance accuracy and compute cost.

Actual Tip Geometry and Machine Learning for Shallow Indentation | Nov 2023 - 2026

- Incorporated microscopy-based spherical tip metrology into FEM by using measured tip profiles instead of ideal radii.
- Reduced experiment-to-simulation mismatch from ~20% to <5%, improving shallow-depth agreement and enabling higher-fidelity property extraction.
- Built a 1,000-case actual-tip 3D FE database to train ML models for rapid property identification (elastic modulus MAPE of 7.6% at 200 nm); published in *J. Mater. Res. Technol.* (2026).

AI-Powered Magnetic Materials Characterization, UTK-MRSEC Hackathon | Mar 2026

- Built an AFM/MFM image-analysis pipeline in ~3 hours (FFT, watershed segmentation, texture maps) showing that magnetic ordering in a thin-film library is composition-driven.

ACADEMIC ACHIEVEMENTS & RECOGNITIONS

- **Jim and Tricia Martin Graduate Fellowship,** University of Tennessee, Knoxville, TN, USA (Oct 2026)
- **Student of the Year,** IRG 2 Graduate Research Assistant (Research Excellence), University of Tennessee, Knoxville, TN, USA (Jul 2025)
- **Best Poster,** IRG 2 Poster Competition, Advanced Materials & Manufacturing, University of Tennessee, Knoxville, TN, USA (Feb 2026)
- **Top-Ranked Scholar (Cohorts 50 and 51), BS in Civil Engineering (Honors Degree)**, Paññāsāstra University of Cambodia, Phnom Penh, Cambodia (Nov 2022)

CONFERENCE PRESENTATIONS

- **USNCCM 18, Chicago, IL (Jul 20 - 24, 2025):** Munireach Nannory, Vivek Chawla, Dayakar Penumadu, Timothy Truster. A novel method for identifying the actual geometry of a spherical nanoindentation tip at shallow depths: Experiments and finite element simulations.
- **ASME IMECE 2024, Portland, OR (Nov 17 - 21, 2024):** Munireach Nannory, Vivek Chawla, Dayakar Penumadu, Timothy Truster. FE nanoindentation simulation for characterization at material extremes. No. IMECE2024 150501.
- **TEAMS 2024, Tennessee Advanced Materials Summit (Jul 9 - 10, 2024):** Munireach Nannory, Vivek Chawla, Dayakar Penumadu, Timothy Truster. Nanoindentation simulations and machine learning for characterization at material extremes.

AWARDS AND FUNDING

- NSF funded PhD Research Assistant, University of Tennessee, Knoxville (Aug 2024 - Present)
- NSF funded MS Research Assistant, University of Tennessee, Knoxville (Aug 2023 - May 2025)
- CAMM Travel Award to present at USNCCM 18, University of Tennessee, Knoxville (Jul 2025)
- Travel Grant, ASME IMECE 2024, Portland, OR (Nov 2024)

UNIVERSITY SEMINARS AND PRESENTATIONS

- **Finite Element Overview and Literature Review for Characterization of Materials at Extremes**, UTK-MRSEC Center for Advanced Materials and Manufacturing, University of Tennessee, Knoxville, TN (2023)
- **Framework for Incorporating Measured Spherical Indenter Geometry in Abaqus/CAE: 2D vs 3D FEM Compared with Experiments**, UTK-MRSEC Center for Advanced Materials and Manufacturing, University of Tennessee, Knoxville, TN (2024)
- **Advancements in Finite Element Simulation for Nanoindentation: A Novel Method for Identifying Actual Indenter Geometry at Shallow Penetration Depths**, UTK-MRSEC Center for Advanced Materials and Manufacturing, University of Tennessee, Knoxville, TN (2025)
- **Substrate Correction of Nanoindentation Depth–Load Response: FEM and Machine Learning Prediction of Bulk/Film Response from Thin-Film Nanoindentation**, University of Tennessee, Knoxville, TN (2025)

Poster Presentations (Selected)

- **Best Poster (IRG2),** Learning Substrate-Effect Regimes in Thin-Film Indentation: Depth-Normalized Metrics, Sensitivity Analysis, and Active Experiment Design, Advanced Materials & Manufacturing, UTK (2026)
- Multiscale 2D/3D Finite Element Simulation for Nanoindentation: Mesh, Friction, and Surrogate/ML Modeling, Advanced Materials & Manufacturing, UTK (2025)
- FE Nanoindentation Simulations for Characterization at Material Extremes, Advanced Materials & Manufacturing, UTK (2023)

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed Journal Articles *(* corresponding author)*

- **Munireach Nannory***, Vivek Chawla, Carleton O'Neil, Dayakar Penumadu, Timothy Truster, "Rapid Mechanical Property Identification from Shallow Nanoindentation Using Actual Tip Geometry and Machine Learning." *Journal of Materials Research and Technology* (IF 7.3), 2026, in press. DOI: 10.1016/j.jmrt.2026.10.191
- Vivek Chawla, Stephen B. Puplampu, **Munireach Nannory**, Chloe Kirk, Sergei Kalinin, Dayakar Penumadu, Timothy Truster, "Towards Plastic Property Extraction from Shallow Indentation Data." *Materials & Design* (IF 8.2), 271, 117138, 2026. DOI: 10.1016/j.matdes.2026.117138

Manuscripts in Preparation

- **Munireach Nannory*** et al., "Governed Autonomous Nanoindentation on a Commercial Instrument with LLM-Orchestrated Multi-Agent Control." **In preparation.** *(LLM multi-agent control of a commercial nanoindenter; ~3 h autonomous hardware campaign.)*
- **Munireach Nannory*** et al., "A FEM and Machine Learning Framework for Substrate-Effect Correction in Thin-Film Nanoindentation." **Full draft complete.** *(1D CNN; <2% MAE; ~60–80% experimental error reduction.)*
- **Munireach Nannory*** et al., "Learning Substrate-Effect Regimes in Thin-Film Indentation: Depth-Normalized Metrics, Sensitivity Analysis, and Active Experiment Design." **In preparation.** *(112,200 simulations; regime maps; inverse planner for target error tolerances.)*